



ADITYA UNIVERSITY

Data Analysis Essentials Cornerstone Project

First Review

Student Performance Data Analysis System

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The **Student Performance Data Analysis System** is a Python-based data analytics project that helps analyze student academic records and understand the factors affecting academic performance. The project uses the **Pandas** library to load and process student datasets, performs data cleaning to handle missing or inconsistent values, calculates average marks and grades, analyzes attendance records, and visualizes performance trends using **Matplotlib** charts. It demonstrates a complete data analysis workflow, enabling users to gain meaningful insights into student performance and providing a strong foundation for future Data Science and Machine Learning applications.

Educational institutions manage the academic records of many students.

Student data includes:

- Student Name
- Roll Number
- Attendance
- Subject Marks
- Average Marks
- Grade
- Subject-wise Performance

Manual analysis is:

- Time consuming
 - Error prone
 - Difficult for large datasets
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- Using **Python Data Analysis**, we can automatically clean, analyze and visualize the data to assist Students Data.

Existing System:

Student performance records are generally maintained manually.

Data is usually stored in:

Excel Sheets

CSV Files

Paper Records

Problems include:

Slow processing

Human errors

Duplicate records

Difficult to identify performance trends

No automatic visualization

Difficult to compare student performance

Limitations of Existing System:

- Large amount of manual work
- Missing data affects decision making
- Difficult to compare applicants
- No graphical insights Poor prediction accuracy

Proposed System :

Develop a **Student Performance Data Analysis System** using Python.

The system performs:

- Load dataset using Pandas
- Explore the dataset
- Clean missing or duplicate data
- Calculate average marks
- Assign grades automatically
- Generate Graphs and visualizations
- Analyze attendance records
- Identify top-performing and weak-performing students
- Generate performance reports

Advantages

- Faster data processing
- Better data quality
- Automatic cleaning
- Easy visualization
- Better understanding of customer profiles

Feature	Description
Student_ID	Unique Student Number
Name	Student Name
Gender	Male/Female
Age	Student Age
Maths	Maths Marks
Physics	Physics Marks
Chemistry	Chemistry Marks
English	English
Computer_Science	Computer Science Marks
Attendance	Attendance Percentage
Average	Average Marks
Grade	Student Grade

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 20 GB
- Input Devices: Keyboard, Mouse
- Output Device: Monitor

Software Requirements

- Operating System: Windows / Linux
- Programming Language: python
- IDE: Google Collab/ Jupyter Notebook
- Libraries: Pandas, Numpy, Matplotlib
- GitHub (for version control)

Modules:

Module 1

- Dataset Loading
- Import libraries
- Read CSV file
- Load student dataset
- Display first rows

Module 2

- Data Exploration
- Dataset shape
- Column names
- Data types
- Missing values
- Statistical summary

Module 3

- Data Cleaning
- Fill missing values
- Remove duplicates records
- Handle incorrect data

Module 4

Data Preprocessing

- Calculate average marks
- Assign grades
- Sort student records
- Filter student performance

Module 5

Exploratory Data Analysis

Visualize

- Subject-wise average marks
- Attendance Analysis
- Grade distribution

Module 6

Performance Analysis

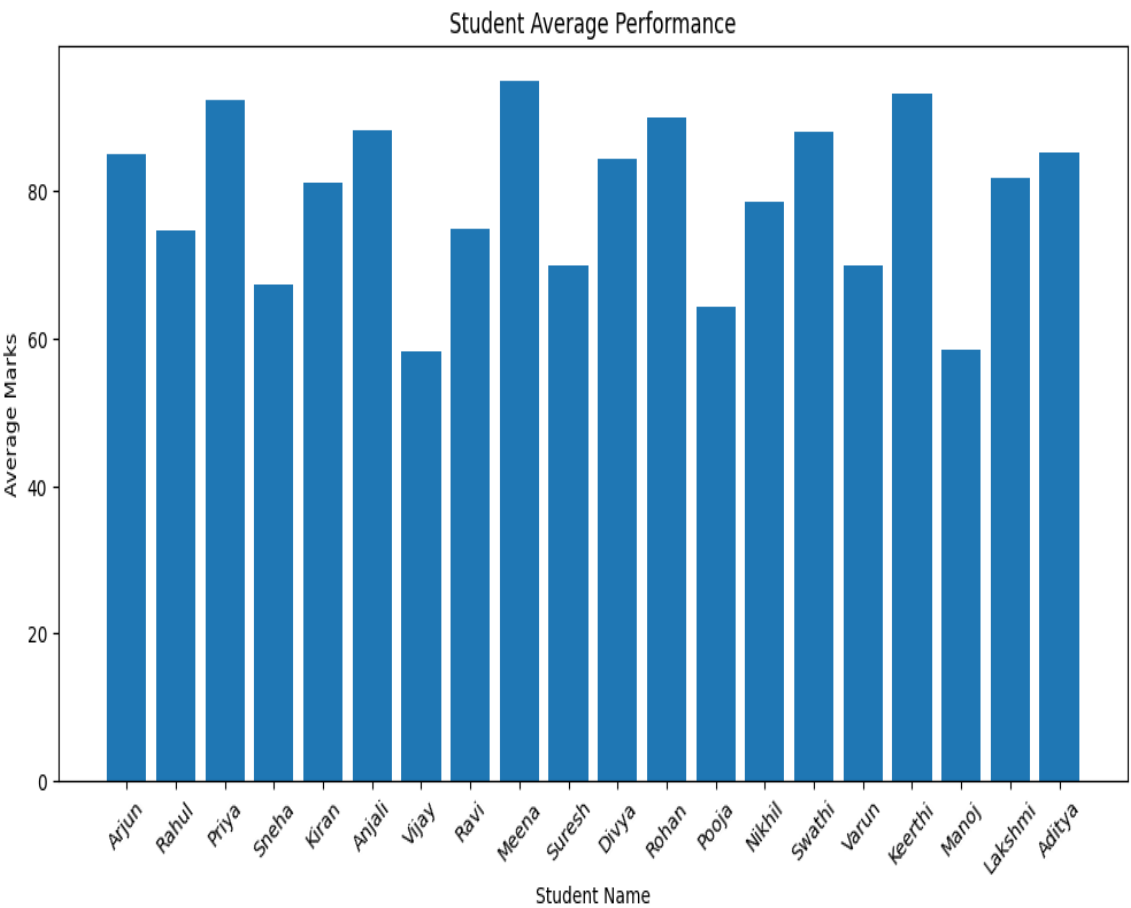
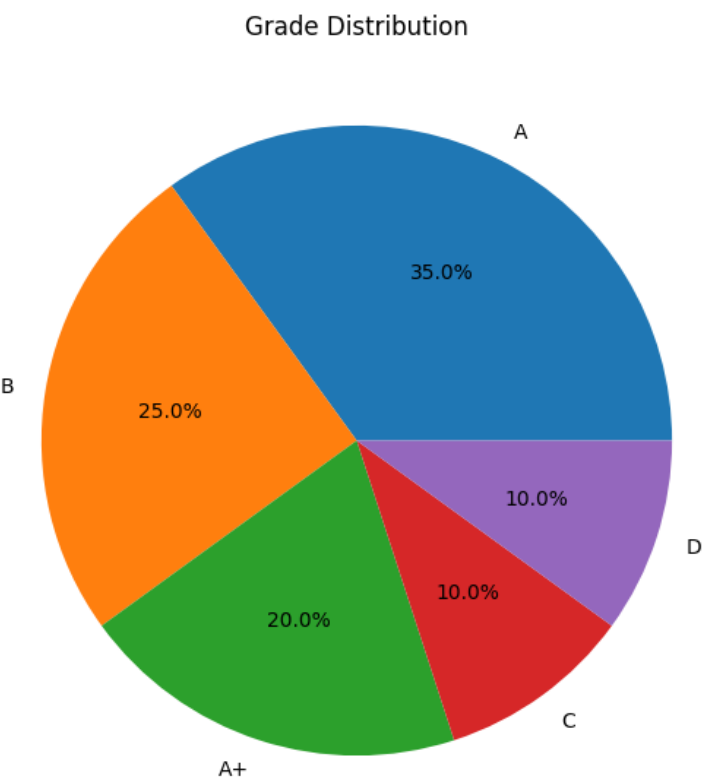
- Top-performing students
- Weak-performing students
- Best-performing student
- Attendance impact on performance

The system displays:

- Dataset Information
- Missing Value Report
- Cleaned Dataset
- Student Performance Report
- Average Marks
- Grade Assignment
- Attendance Analysis
- Charts and Graphs
- Subject-wise Performance
- Topper Information
- Performance Insights

Expected Outputs

Visual Data:



CONCLUSION

The **Student Performance Data Analysis System** demonstrates the complete data analysis pipeline using Python. The project helps students understand:

- The project helps students understand
 - Data Collection
 - Data Cleaning
 - Data Analysis
 - Data Visualization
 - Performance Evaluation
 - Report Generation
- It provides valuable insights into student academic performance and creates a strong foundation for future **Data Science, Machine Learning, and Educational Analytics** projects.

THANK YOU